

Electronic Design and Fabrication with gEDA

Learn how to design and fabricate cool electronic gadgets.

*H*ave you ever wondered how electronic gadgets are designed? The heart of these gadgets is a certain circuit board(s) in a nice shiny cabinet. But how does one design these circuit boards? Resistors, capacitors, transistors, etc., are all brought together in a circuit board to perform a greater function. This article gives a brief description about the tools for electronic design in GNU/Linux.

What is EDA?

EDA (Electronic Design Automation) comprises a set of tools for electronic design. EDA applications are used by circuit designers to design and produce electronic systems. EDA tools are for low- to medium-complexity printed circuit board (PCB) design. A basic EDA tool takes care of the following functionalities:

- Schematic capture
- Netlist generation
- Simulation
- Printed circuit board (PCB) layout
- PCB testing
- PCB fabrication

What is gEDA?

Although most popular EDA tools are proprietary in nature, in this article we'll introduce you to gEDA that is basically a set of GPL'd tools for EDA design, and these applications are collectively referred to as the gEDA Suite. gEDA is pronounced 'g-daahhh' or 'gee-daahhh'. The gEDA project is also known as gaf (for 'gschem and friends').

The gEDA project was started by Ales Hvezda on April 1, 1998, and it included a schematic capture program and a netlister. At present, this project features a complete suite of free software applications for electronics design, which includes: schematic capture, a bulk attribute editor, a bill of materials (BOM) file generator, a design rule check (DRC), a 2D file format netlister, an analogue and digital simulator, and a printed circuit board (PCB) designer. Refer to Tables 1 and 2 for tools that are part of gEDA.

That's a huge list, wouldn't you agree? But what's the use of so many independent tools? Usually, hardware designs are not done using a single piece of software. A series of programs are used for a good hardware design. For instance, in the case of PCB